

Two-Tiered Clinical & Transcriptomic Feature Selection Report

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This report presents a comprehensive **two-tiered feature selection architecture** evaluating prognostic and predictive clinical markers across melanoma patient populations:

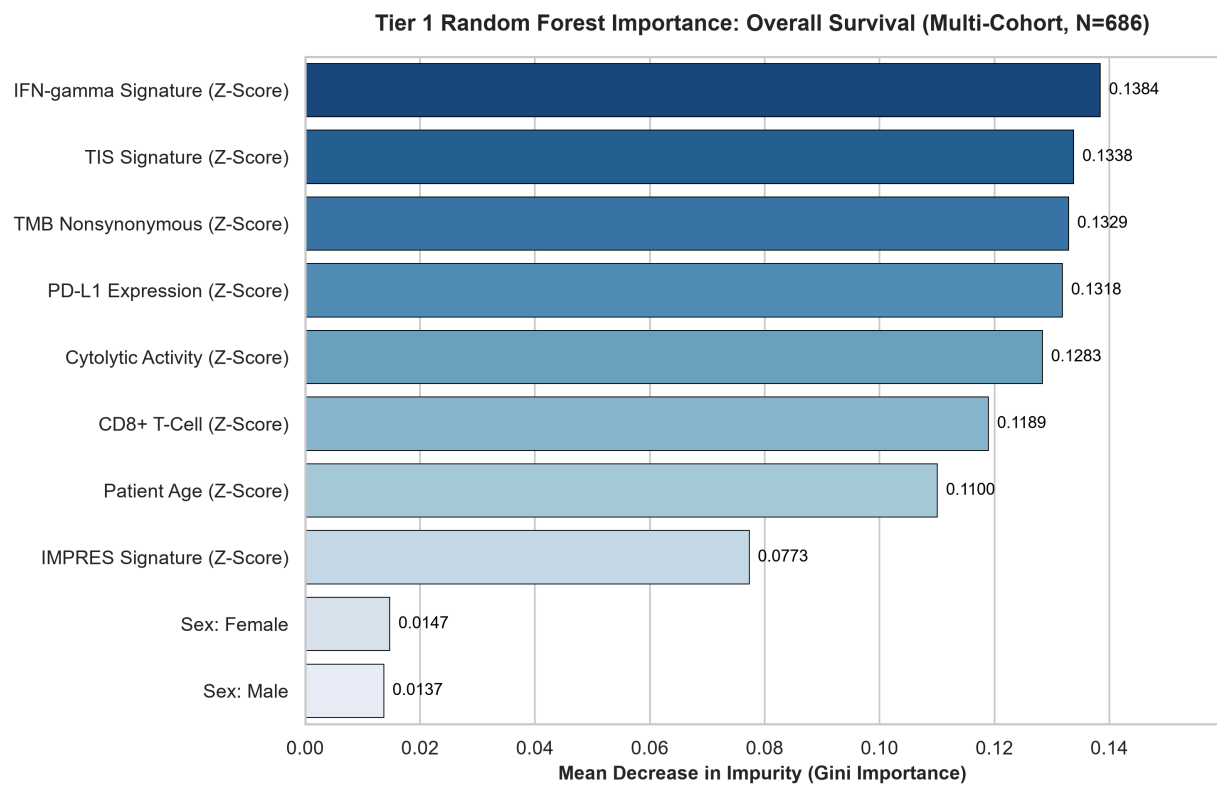
- **Tier 1 (Multi-Cohort Consensus, $N = 699$):** Evaluates 10 harmonized cross-cohort features (6 transcriptomic immune signatures, TMB, age, sex) pooled across all four study cohorts (**TCGA-SKCM**, **Liu 2019**, **Hugo 2016**, **Riaz 2017**).
- **Tier 2 (Granular TCGA Pathological Staging, $N = 443$):** Evaluates 77 detailed clinical, pathological TNM staging, anatomical site, aneuploidy, and hypoxia attributes specifically within the **TCGA-SKCM** reference cohort.

1. Tier 1: Multi-Cohort Feature Selection (N=699)

- **Total Merged Sample Size:** 699 patients across 4 cohorts
- **Overall Survival Evaluation Cohort:** 686 patients
- **Cox Survival Evaluation Cohort:** 676 patients
- **Anti-PD-1 Response Evaluation Cohort:** 247 trial patients
- **FDR-Significant Survival Predictors ($FDR < 0.05$):** 7 features

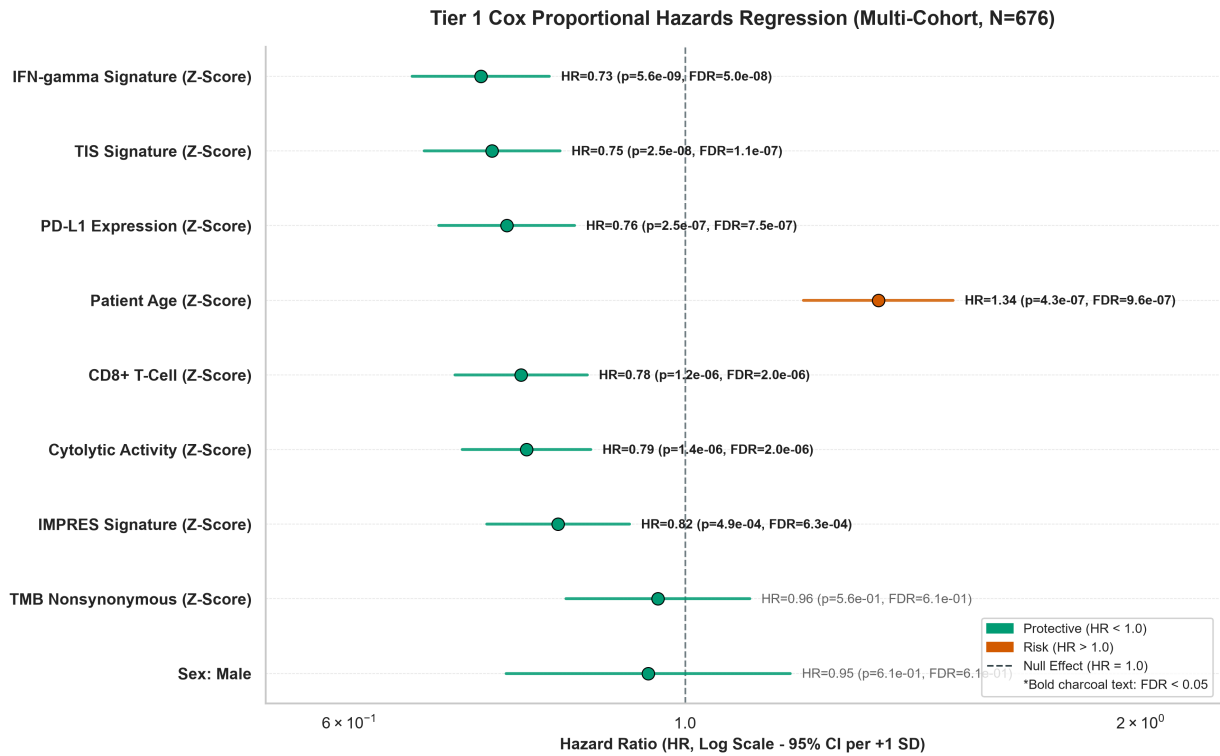
1.1. Random Forest Importance for Overall Survival (N=686)

Random Forest feature importance (500 estimators) trained on the $N = 686$ overall survival cohort:



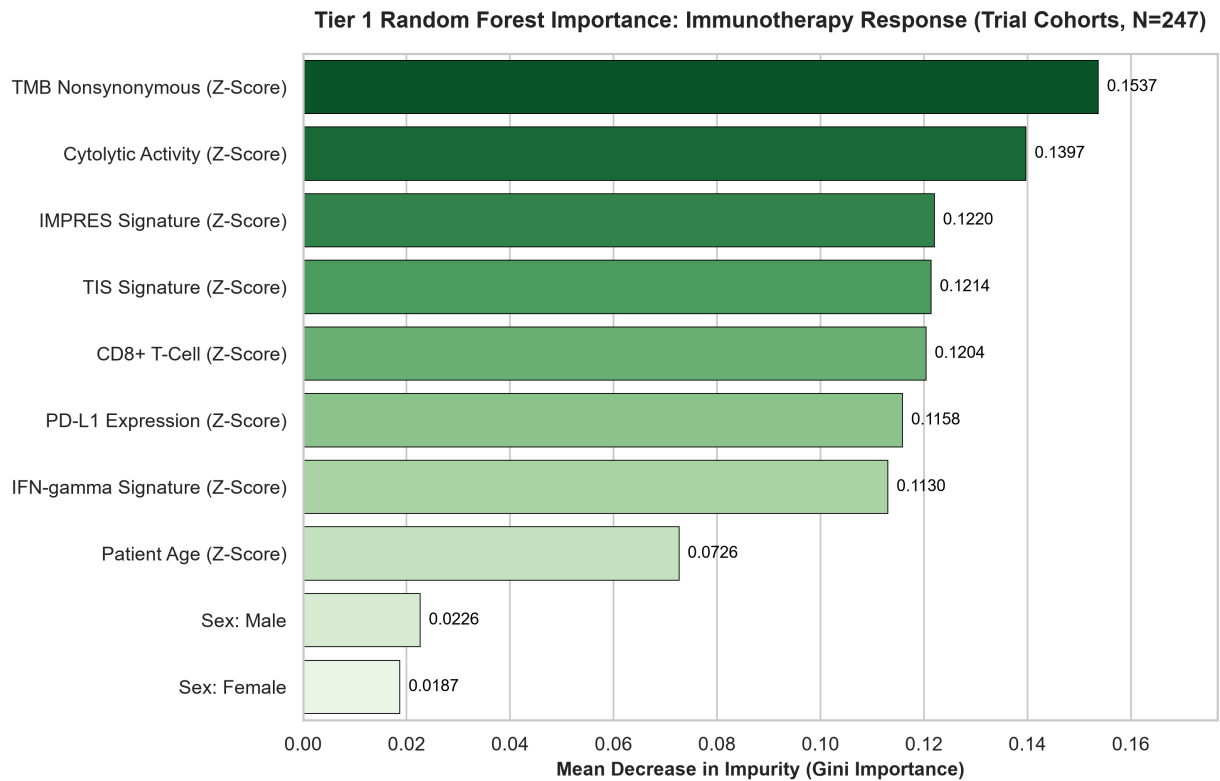
1.2. Univariate Cox Proportional Hazards Regression (N=676)

Univariate Cox Proportional Hazards models fitted across $N = 676$ patients with complete survival duration data:



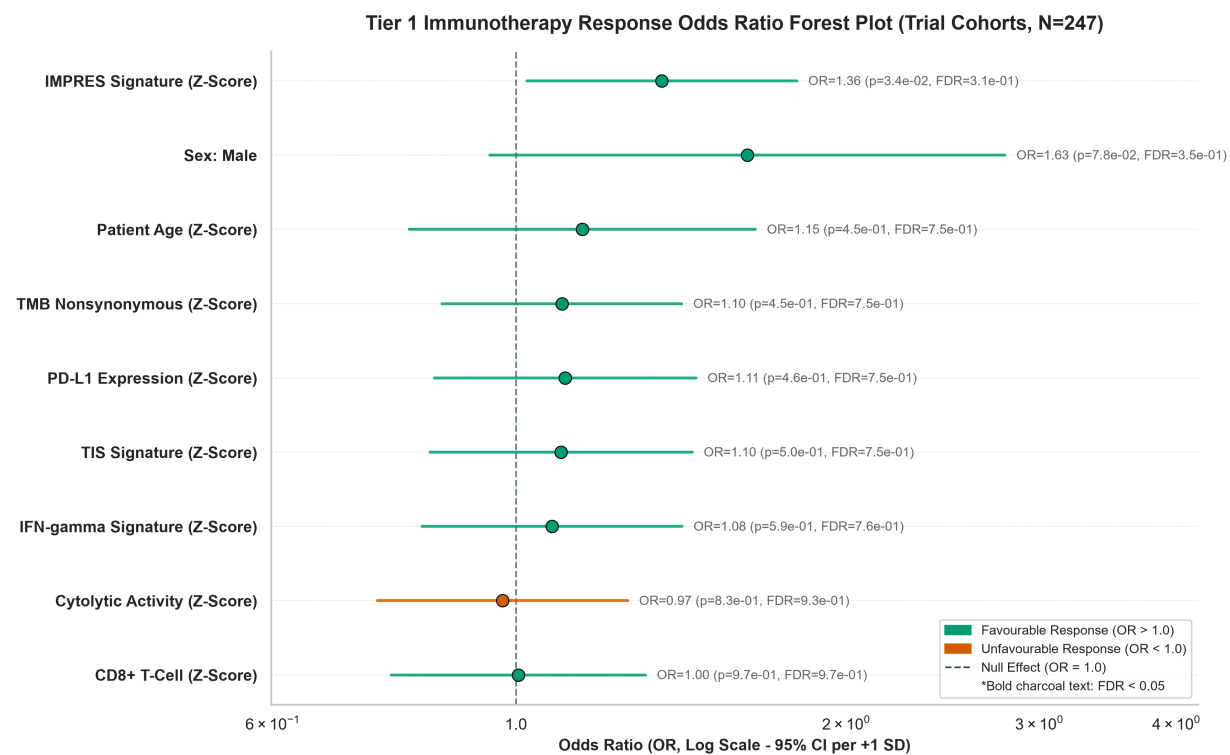
1.3. Random Forest Importance for Anti-PD-1 Immunotherapy Response (N=247)

Random Forest feature importance predicting objective response (CR/PR vs PD) across the $N = 247$ trial cohort:



1.4. Univariate Forest Plot for Anti-PD-1 Immunotherapy Response (N=247)

Univariate Logistic Regression Odds Ratio (OR) forest plot predicting objective anti-PD-1 response across the $N = 247$ trial cohort:



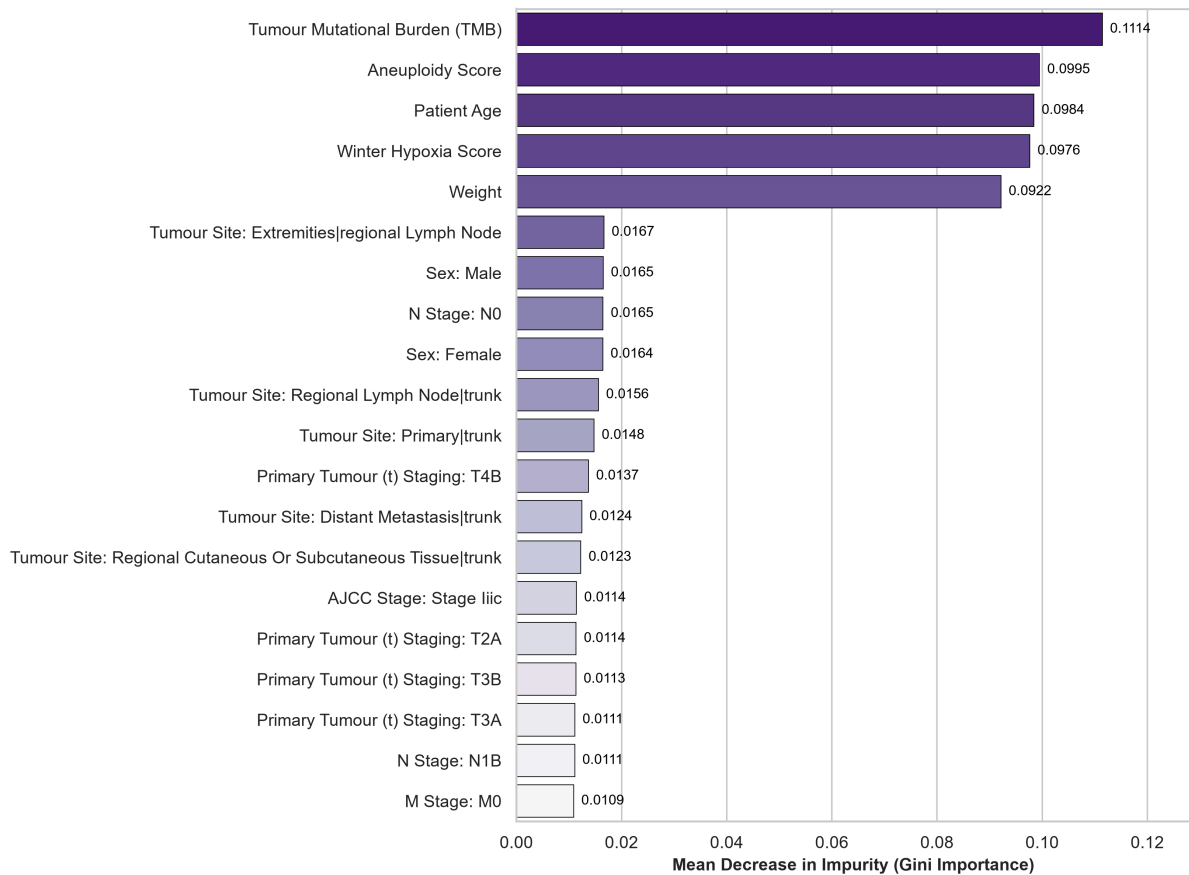
2. Tier 2: Granular TCGA Pathological & Clinical Staging (N=443)

- **TCGA Total Cohort Sample Size:** 443 patients
- **TCGA OS Classification Cohort:** 436 patients
- **TCGA Cox Survival Evaluation Cohort:** 427 patients
- **Encoded Dummy Features:** 77 dummy variables
- **FDR-Significant Pathological Predictors (FDR < 0.05):** 10 features

2.1. Random Forest Importance for Granular TCGA Clinical Attributes (N=436)

Top 20 Random Forest clinical predictors trained on $N = 436$ TCGA patients with non-null survival status:

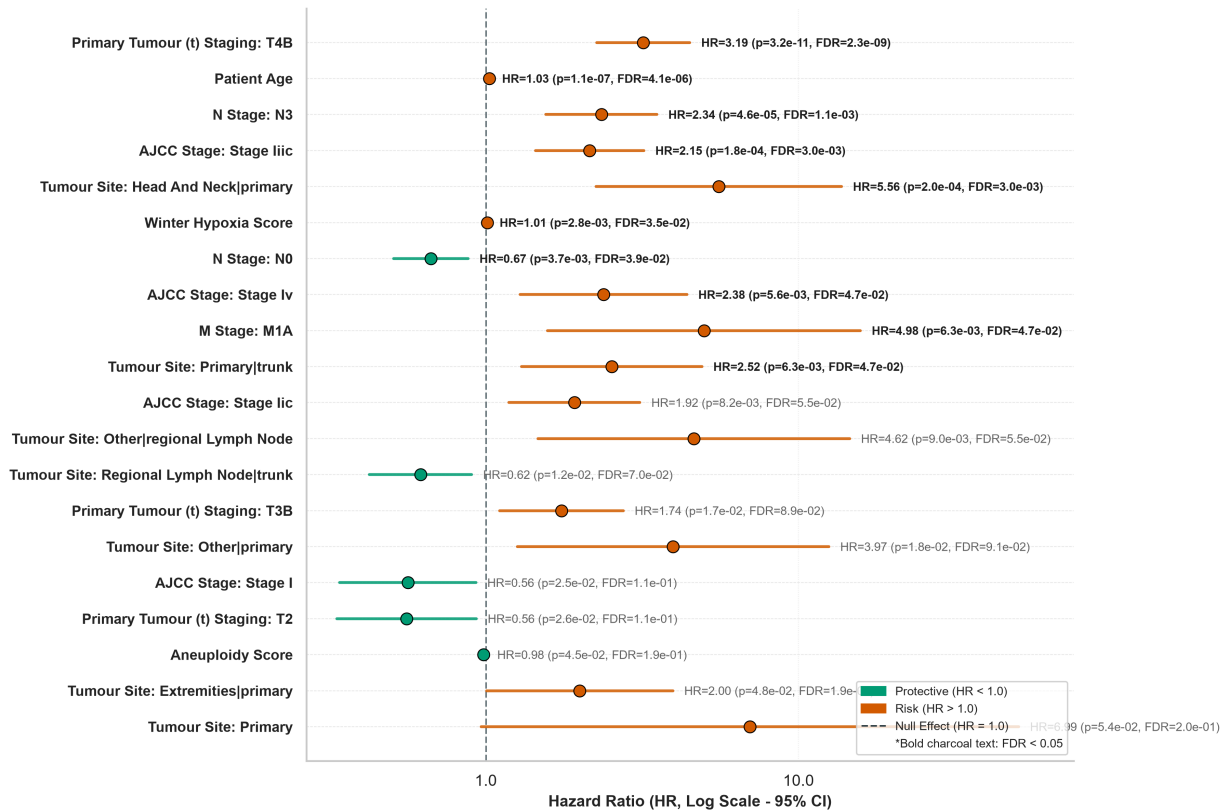
Tier 2 Random Forest Importance: Granular TCGA Clinical Predictors (N=436)



2.2. Univariate Cox Proportional Hazards Regression for TCGA Attributes (N=427)

Top 20 Univariate Cox hazard ratios evaluated across $N = 427$ TCGA patients with complete survival duration data:

Tier 2 Cox Proportional Hazards Regression: Top TCGA Predictors (N=427)



3. Key Analytical & Biological Summary

1. **Tier 1 Top Survival Biomarker:** IFN-gamma Signature (Z-Score) is the single strongest protective statistical predictor across all 4 cohorts (Hazard Ratio = **0.73**, univariate p-value = **5.61e-09**, FDR = **5.05e-08**).
2. **Tier 1 Top Immunotherapy Marker:** TMB Nonsynonymous (Z-Score) is the top predictive feature for objective anti-PD-1 response across trial cohorts (Gini Importance = **0.1537**).
3. **Tier 2 Top Pathological Staging Marker:** Primary Tumour (t) Staging: T4B is the strongest clinical predictor of survival in TCGA (Hazard Ratio = **3.19**, p-value = **3.17e-11**, FDR = **2.35e-09**).
4. **Biological Alignment:** Microenvironmental T-cell inflammation signatures ($text{IFN}-\gamma$, TIS, CYT, CD8, IMPRES) consistently confer significant mortality risk reduction ($text{HR} < 1.0$, $p < 0.01$) across both multi-cohort and single-cohort models.